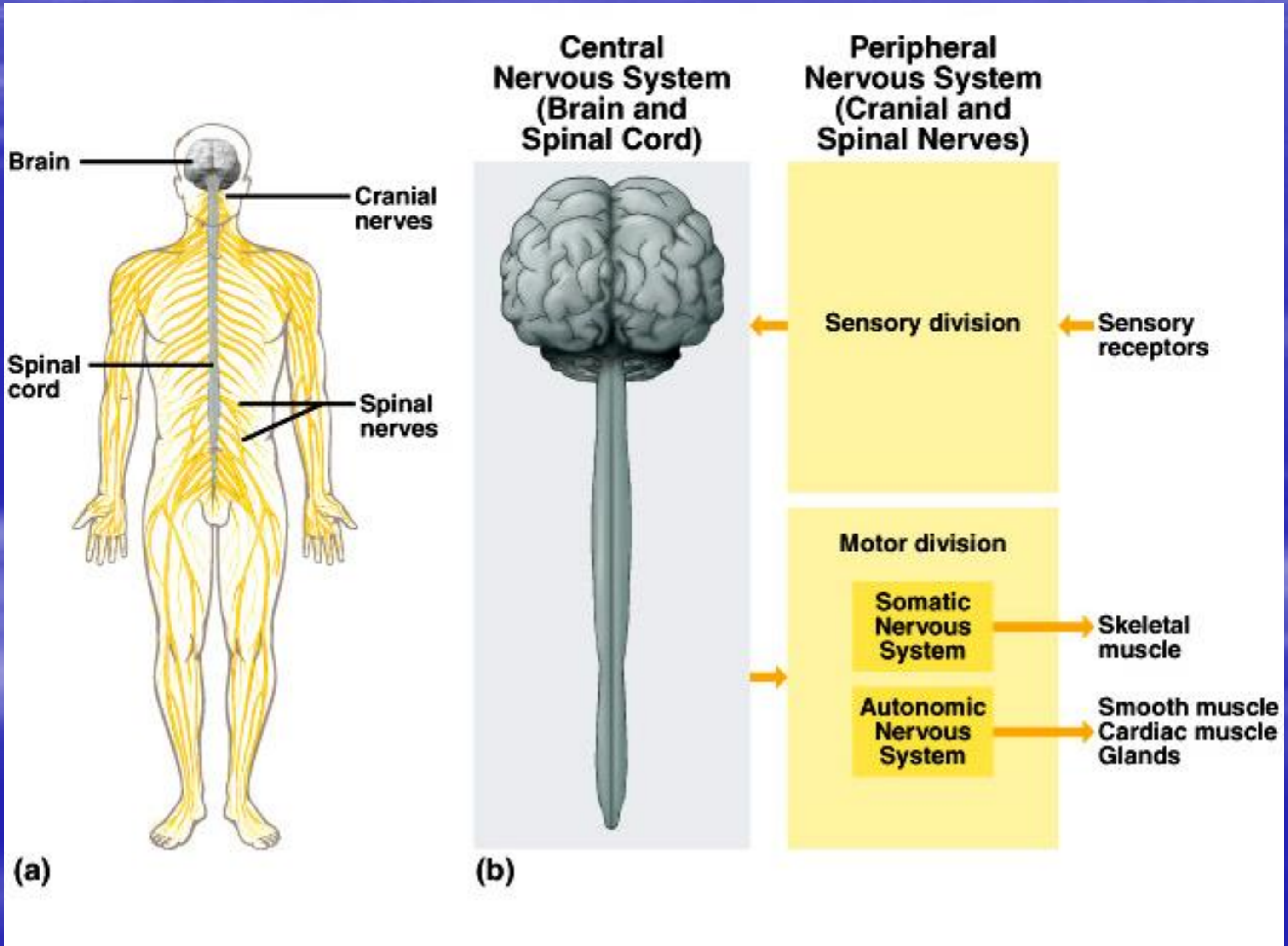
Nervous System

Organization of the Nervous System

- Central nervous system
  1. Brain
  2. Spinal cord

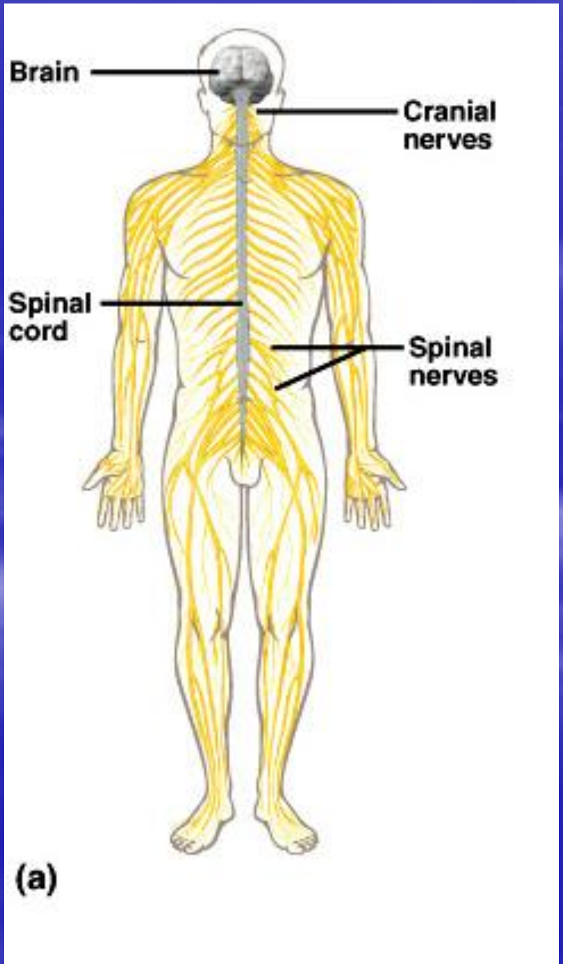
- Peripheral nervous system
  1. Sensory receptors
  2. Sensory nerves
  3. Ganglia

Divisions Nervous System



Divisions of the Nervous System

- Central Nervous System
  - brain
  - spinal cord
- Peripheral Nervous System
  - nerves
    - cranial nerves
    - spinal nerves



Divisions of Peripheral Nervous System

Sensory Division

- **picks up sensory information and delivers it to the CNS**

Motor Division

- **carries information to muscles and glands**

Divisions of the Motor Division

- **Somatic – carries information to skeletal muscle**
- **Autonomic – carries information to smooth muscle, cardiac muscle, and glands**

Functions of Nervous System

Sensory Function

- sensory receptors gather information
- information is carried to the CNS

Integrative Function

- sensory information used to create
  - sensations
  - memory
  - thoughts
  - decisions

Motor Function

- decisions are acted upon
- impulses are carried to effectors

Central Nervous System

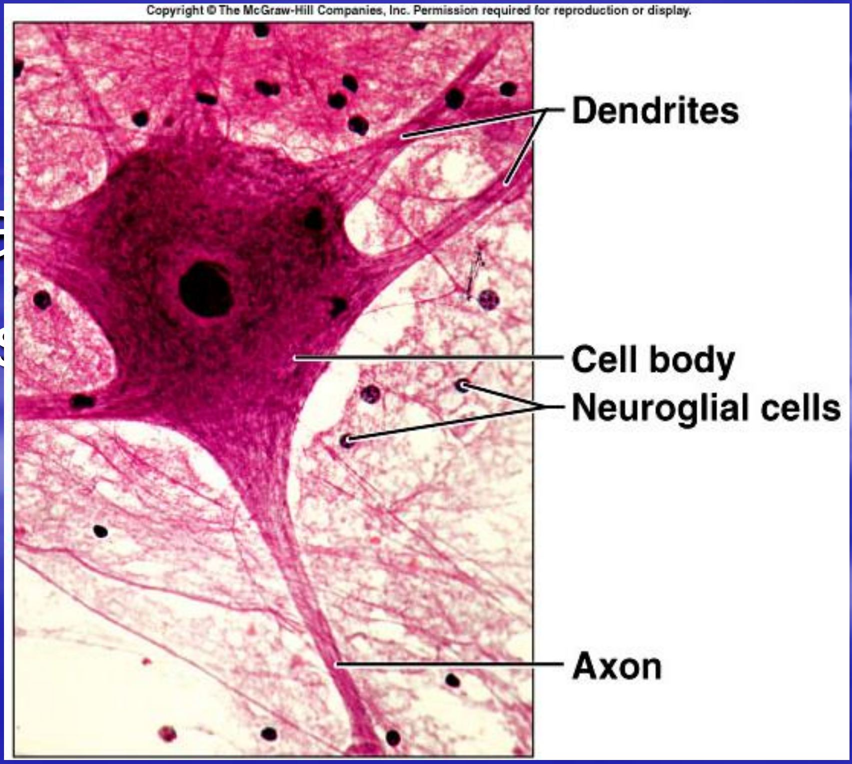
- Functions:
  - Has information about environment
  - Organizes reflexes
  - Plans & execute voluntary movement
  - Memories, thinking & learning

Nerve Tissue

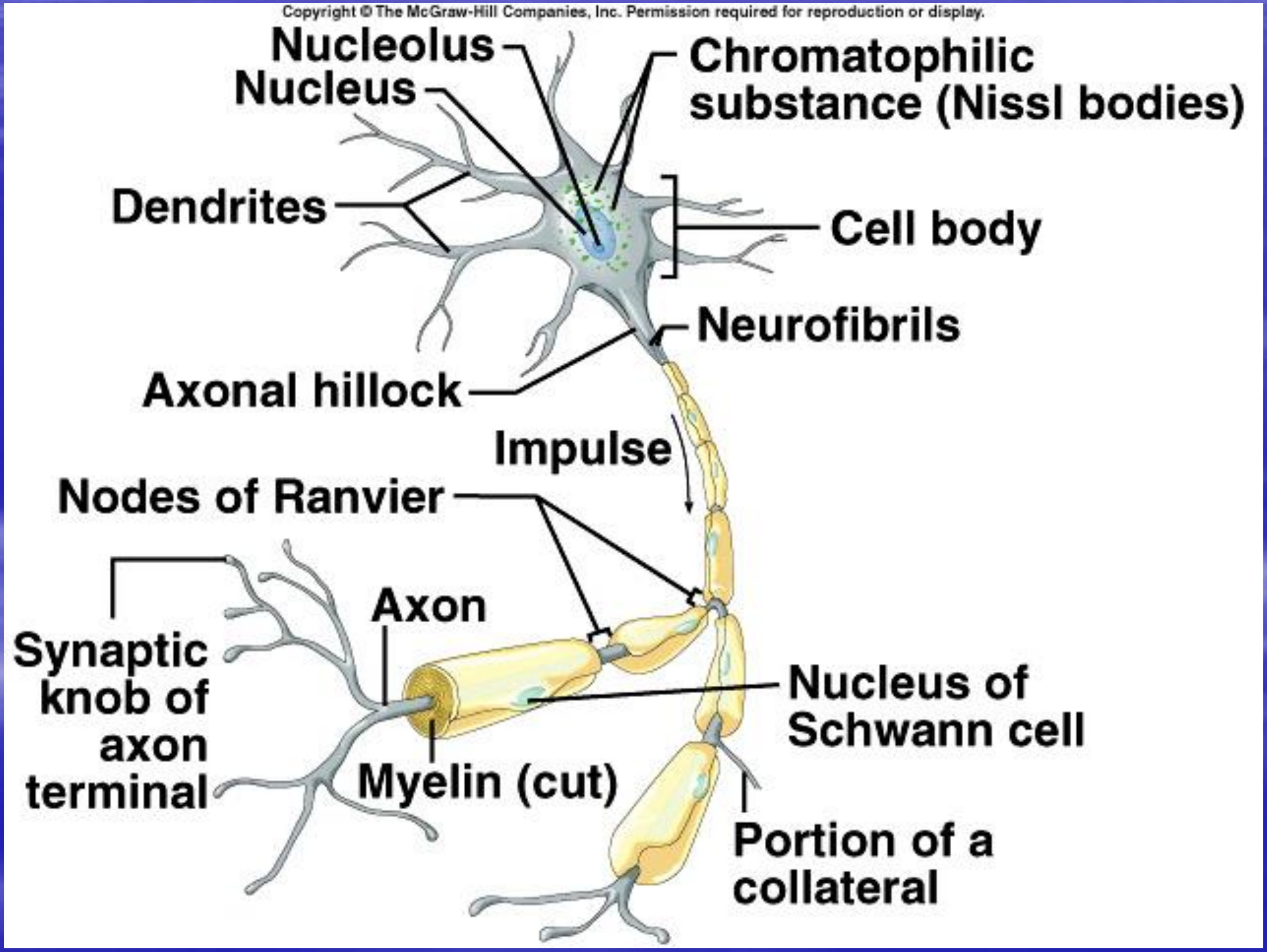
- The two principal cell types of the nervous system are:
- Neurons – excitable cells

that transmit electrical signa

– Neuroglia - supporting cells



Neuron Structure



Properties of neurones

1- irritability:

- **is the ability to initiate nerve impulses in response to stimuli from:**
- **• outside the body, e.g. touch, light waves**
- **• inside the body, e.g. a change in the concentration of carbon dioxide in the blood alters respiration; a thought may result in voluntary movement.**

2- Conductivity: means the ability to transmit an impulse.

Classification of Neurons – Structural Differences

Bipolar

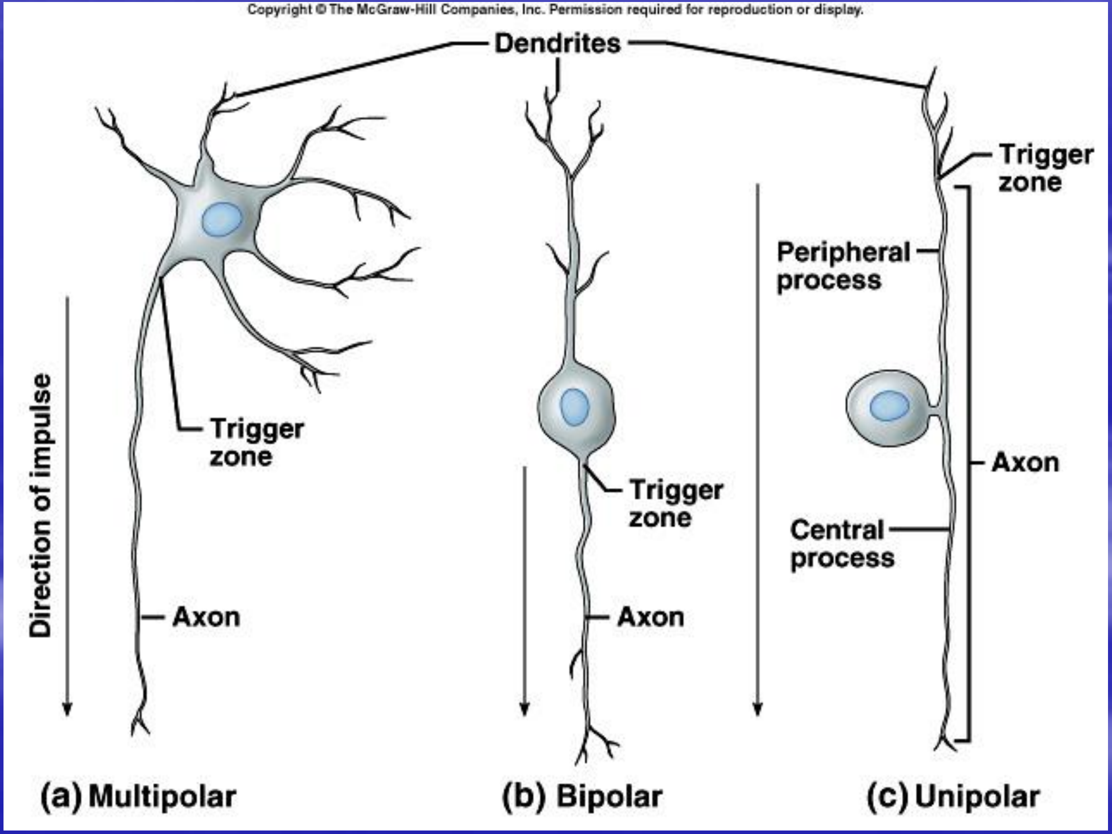
- two processes
- eyes, ears, nose

Unipolar

- one process
- ganglia

Multipolar

- many processes
- most neurons of CNS



Classification of Neurons – Functional Differences

Sensory Neurons

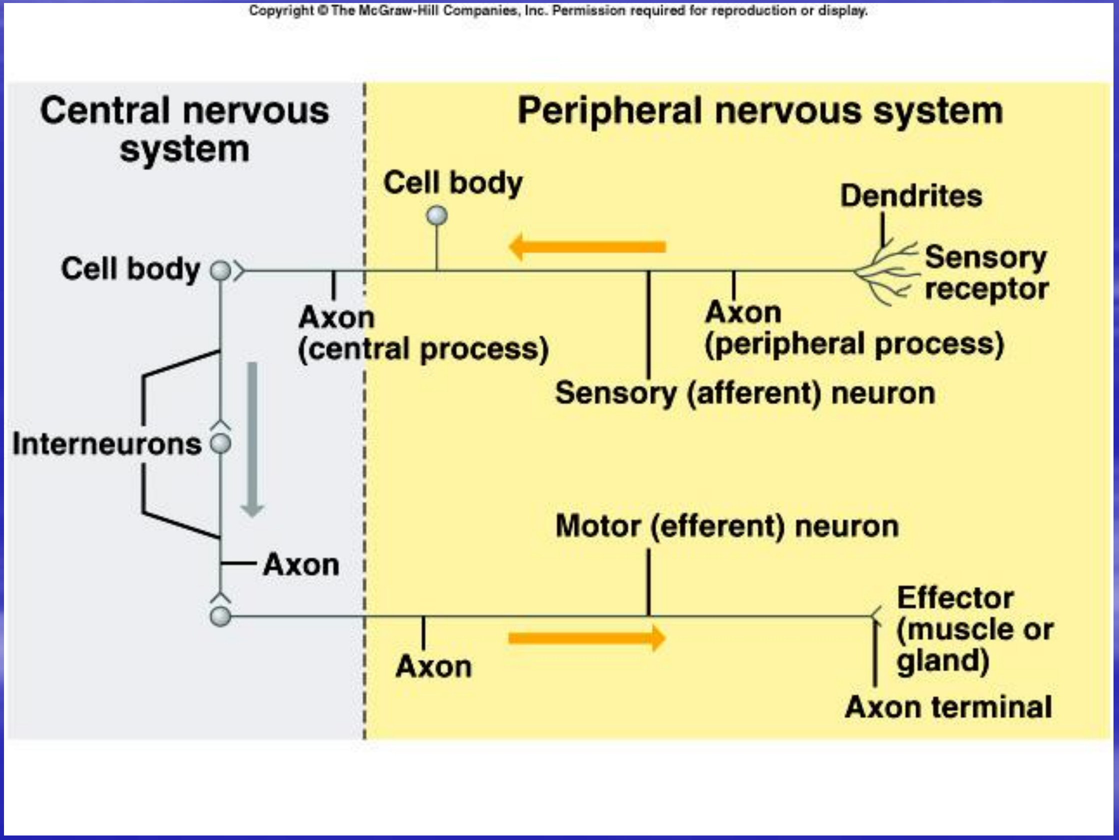
- afferent
- carry impulse to CNS
- most are unipolar
- some are bipolar

Interneurons

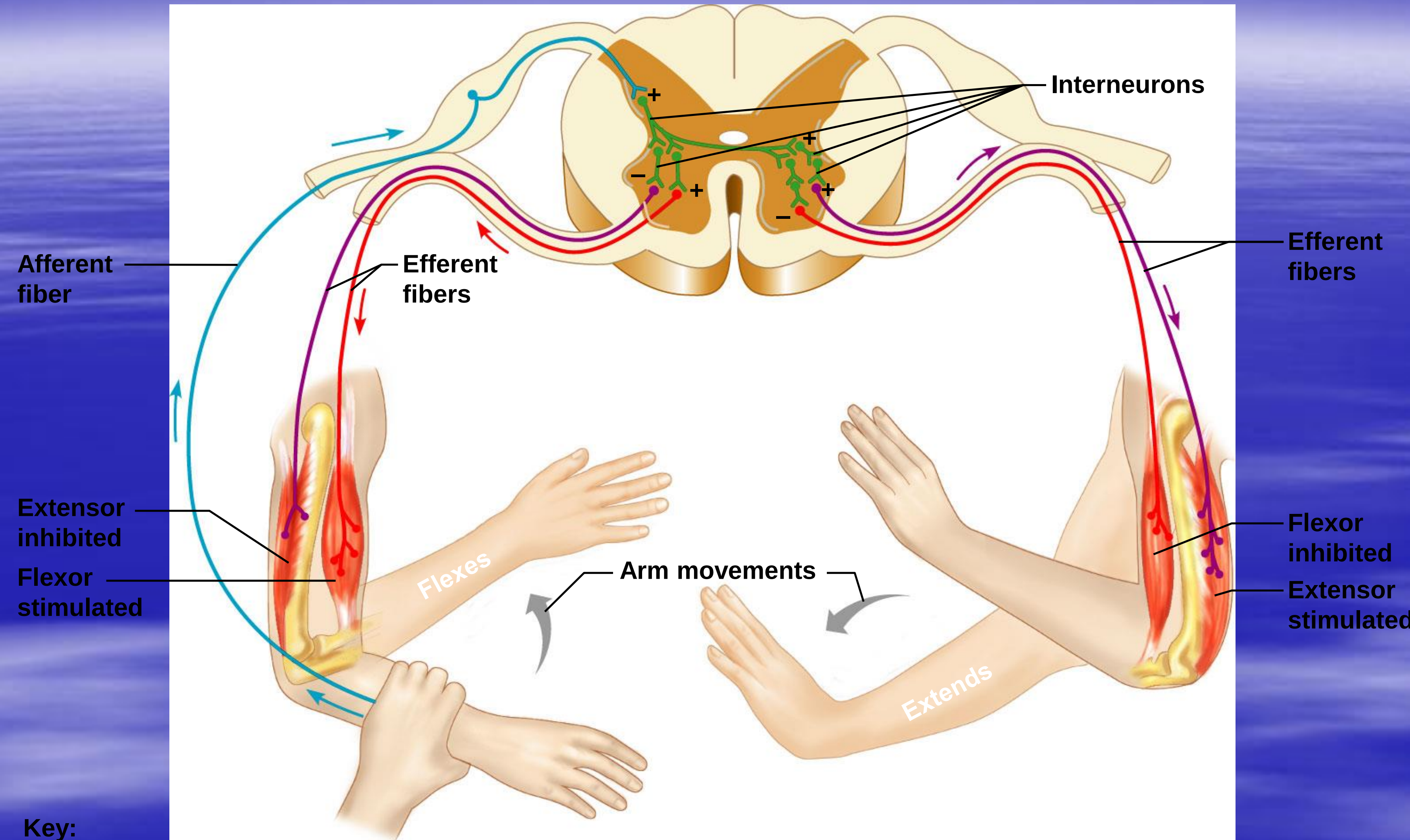
- link neurons
- multipolar
- in CNS

Motor Neurons

- multipolar
- carry impulses away from CNS
- carry impulses to effectors



Classification of neurons by function



Key:
+ Excitatory synapse
– Inhibitory synapse

Right arm (site of stimulus)

Left arm (site of reciprocal activation)

Neuroglia

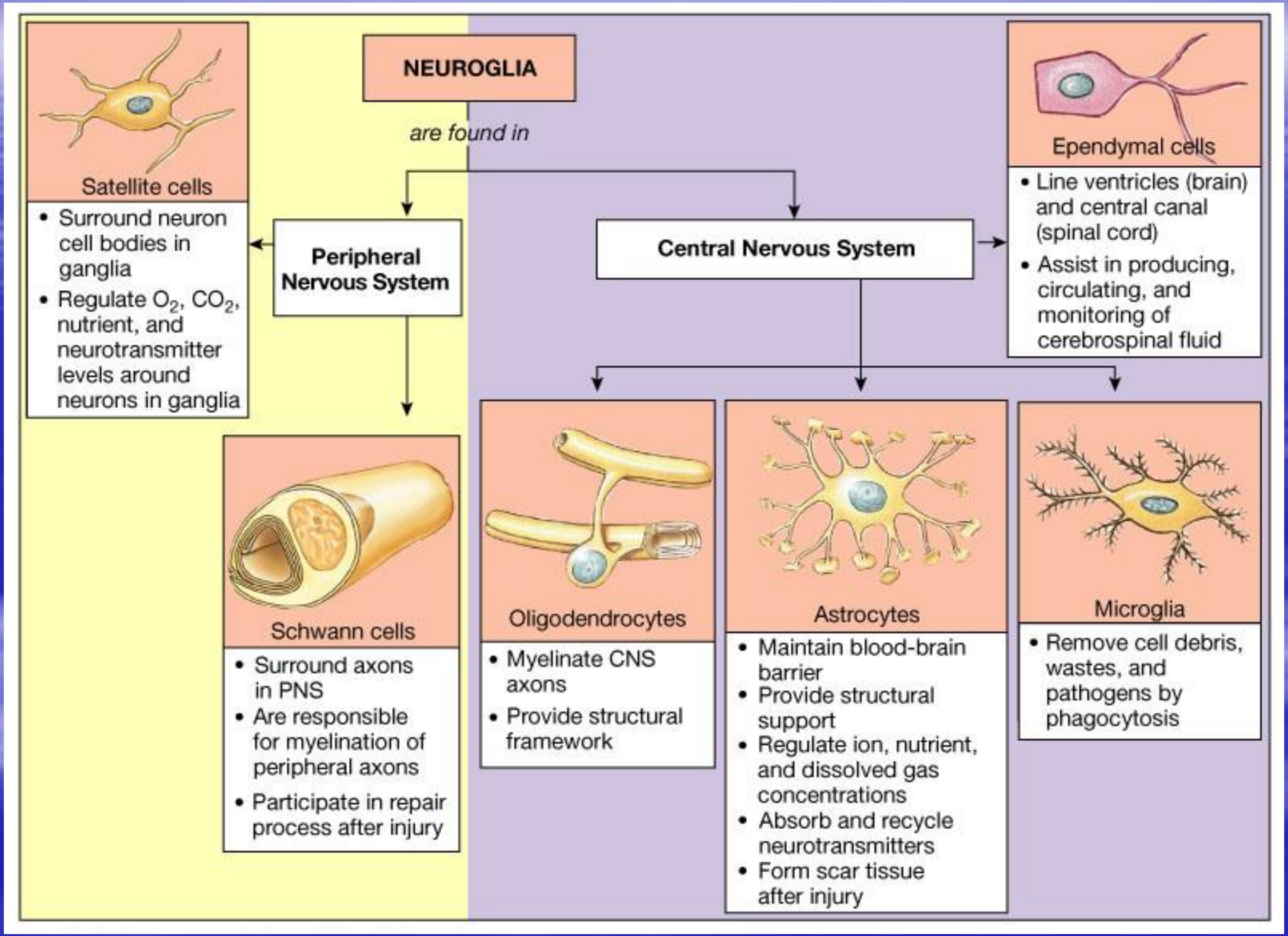
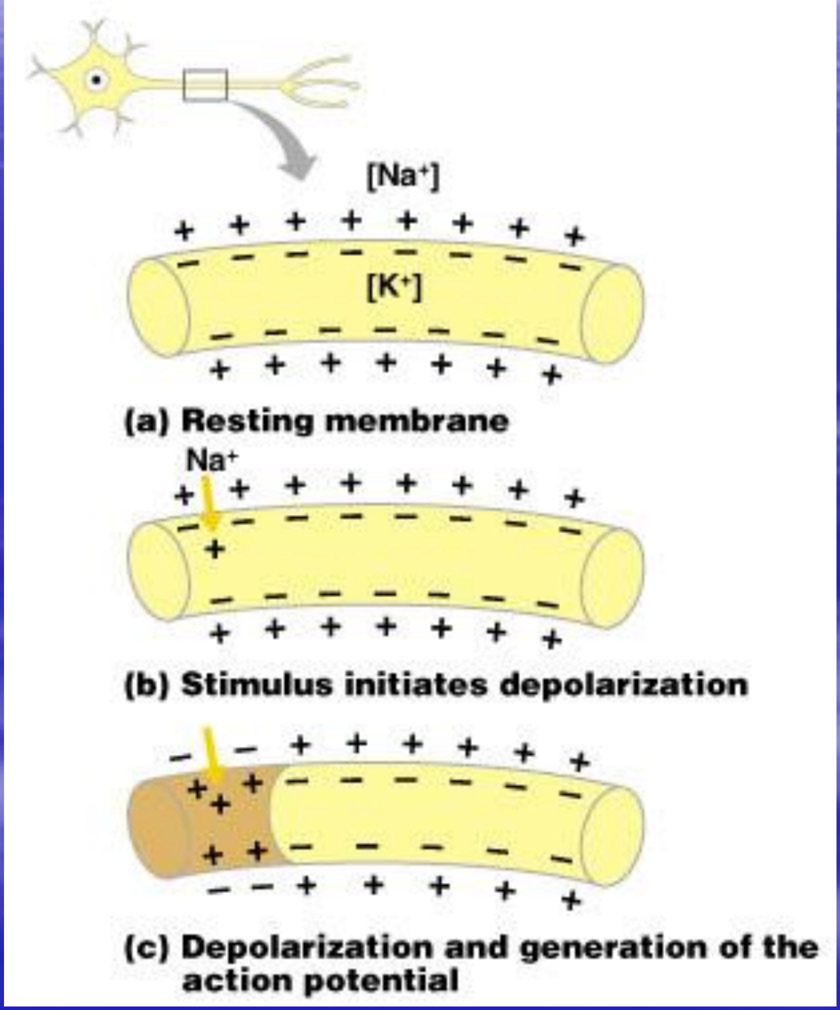


Figure 12.6

Starting a Nerve Impulse

- Depolarization – a stimulus depolarizes the neuron's membrane
- A deploarized membrane allows sodium (Na^+) to flow inside the membrane
- The exchange of ions initiates an action potential in the neuron

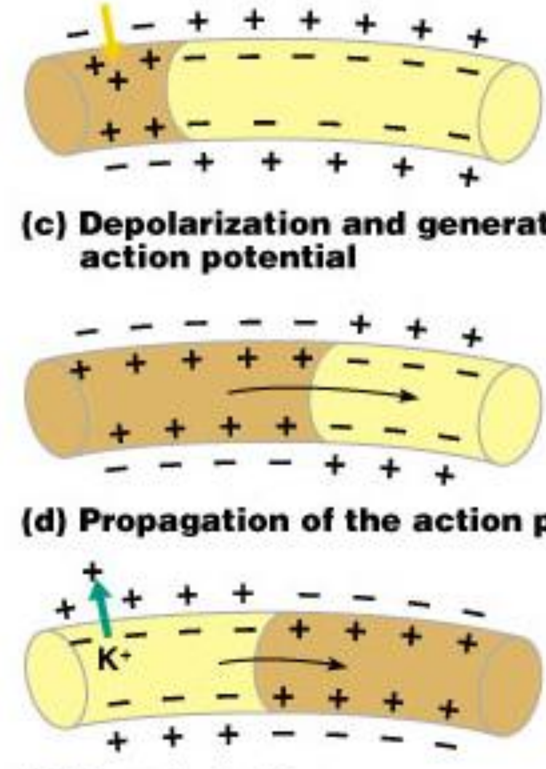


The Action Potential

- If the action potential (nerve impulse) starts, it is propagated over the entire axon
- Potassium ions rush out of the neuron after sodium ions rush in, which repolarizes the membrane
- The sodium-potassium pump restores the original configuration
  - This action requires ATP

Nerve Impulse Propagation

- The impulse continues to move toward the cell body
- Impulses travel faster when fibers have a myelin sheath



(c) Depolarization and generation of the action potential

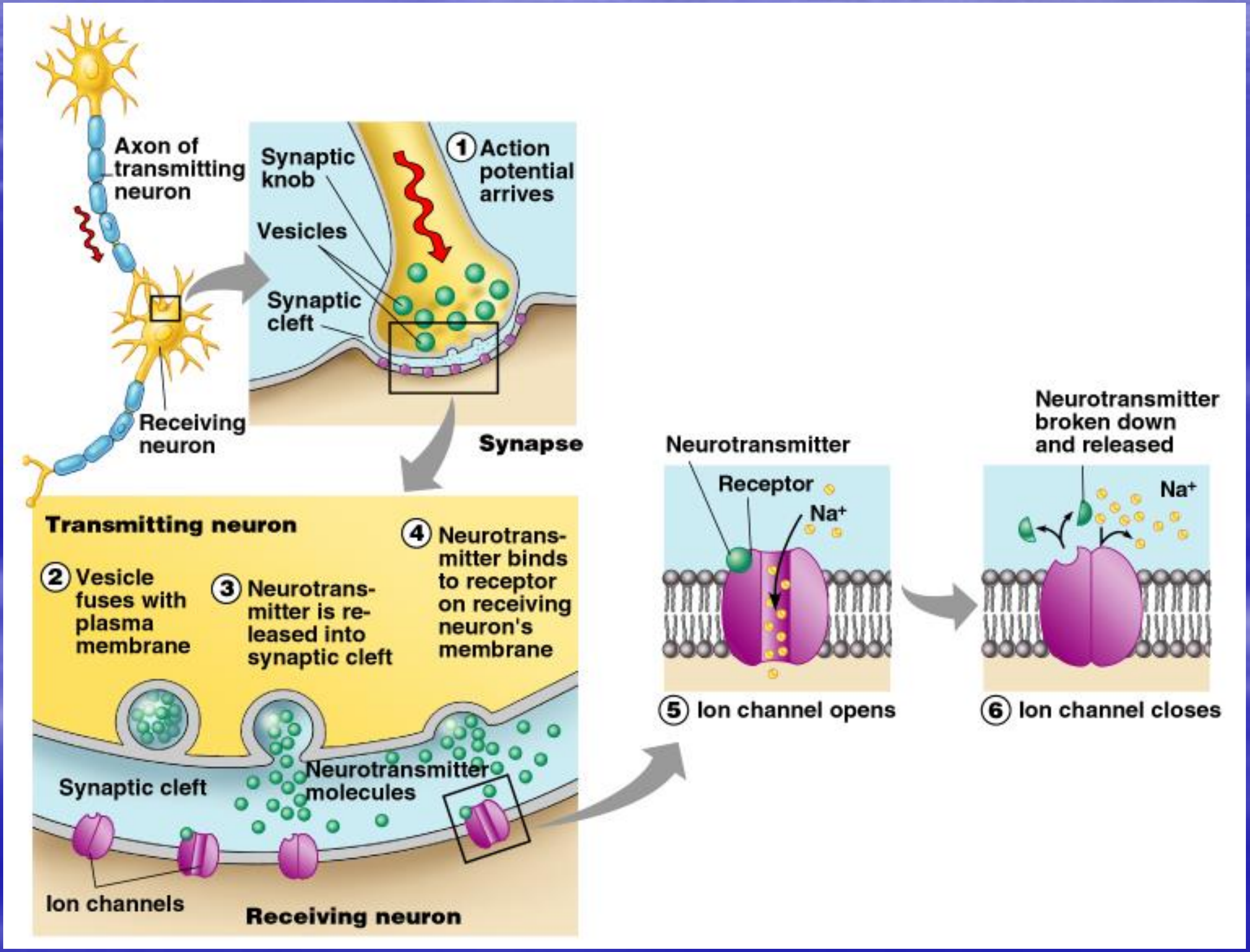
(d) Propagation of the action potential

(e) Repolarization

Continuation of the Nerve Impulse between Neurons

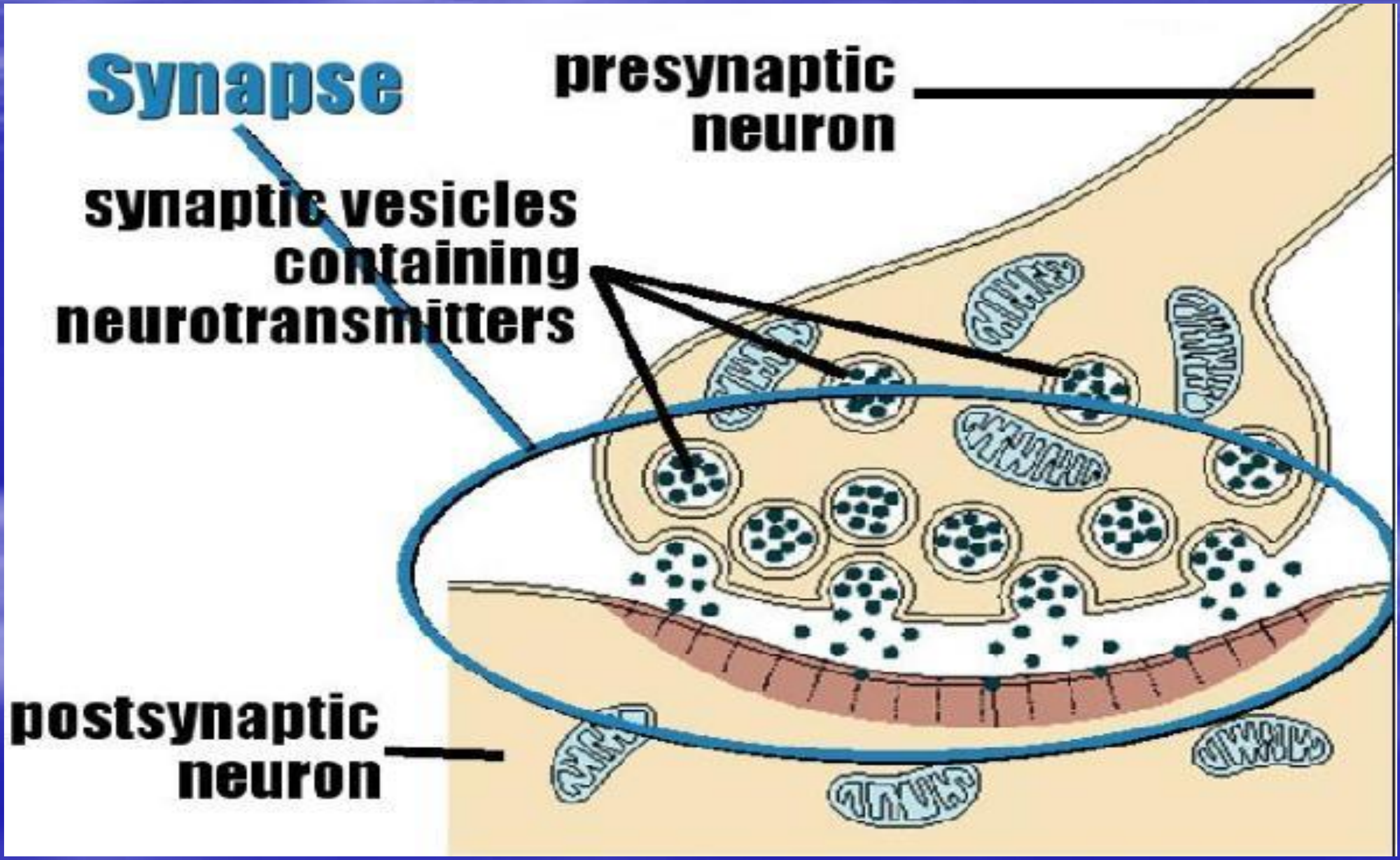
- Impulses are able to cross the synapse to another nerve
  - Neurotransmitter is released from a nerve’s axon terminal
  - The dendrite of the next neuron has receptors that are stimulated by the neurotransmitter
  - An action potential is started in the dendrite

How Neurons Communicate at Synapses



SYNAPSES

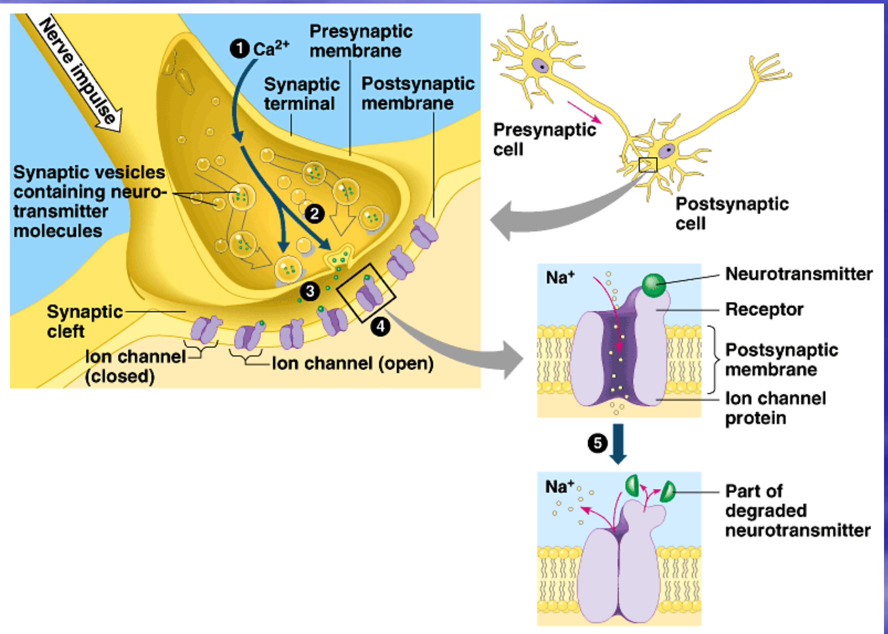
It is an area of contact between two neurons

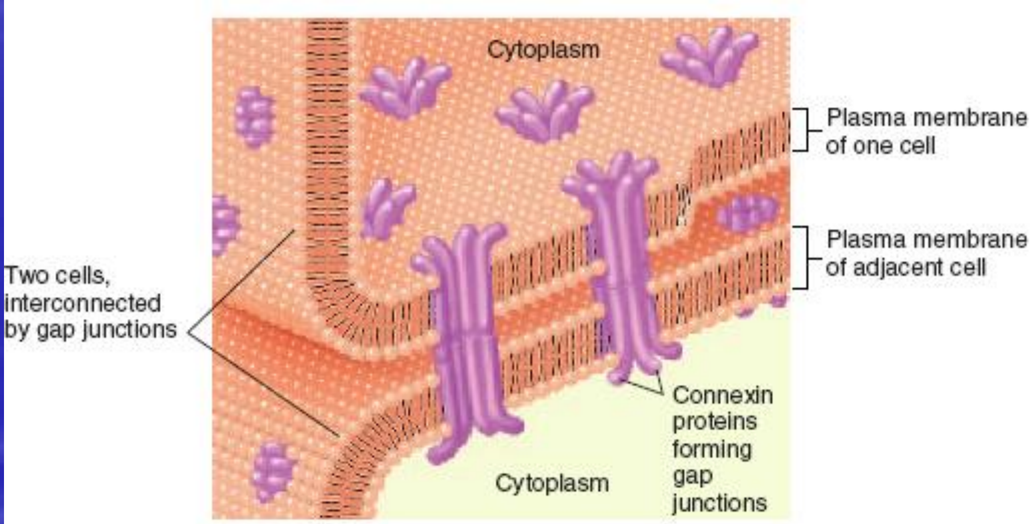


Types of Synapses

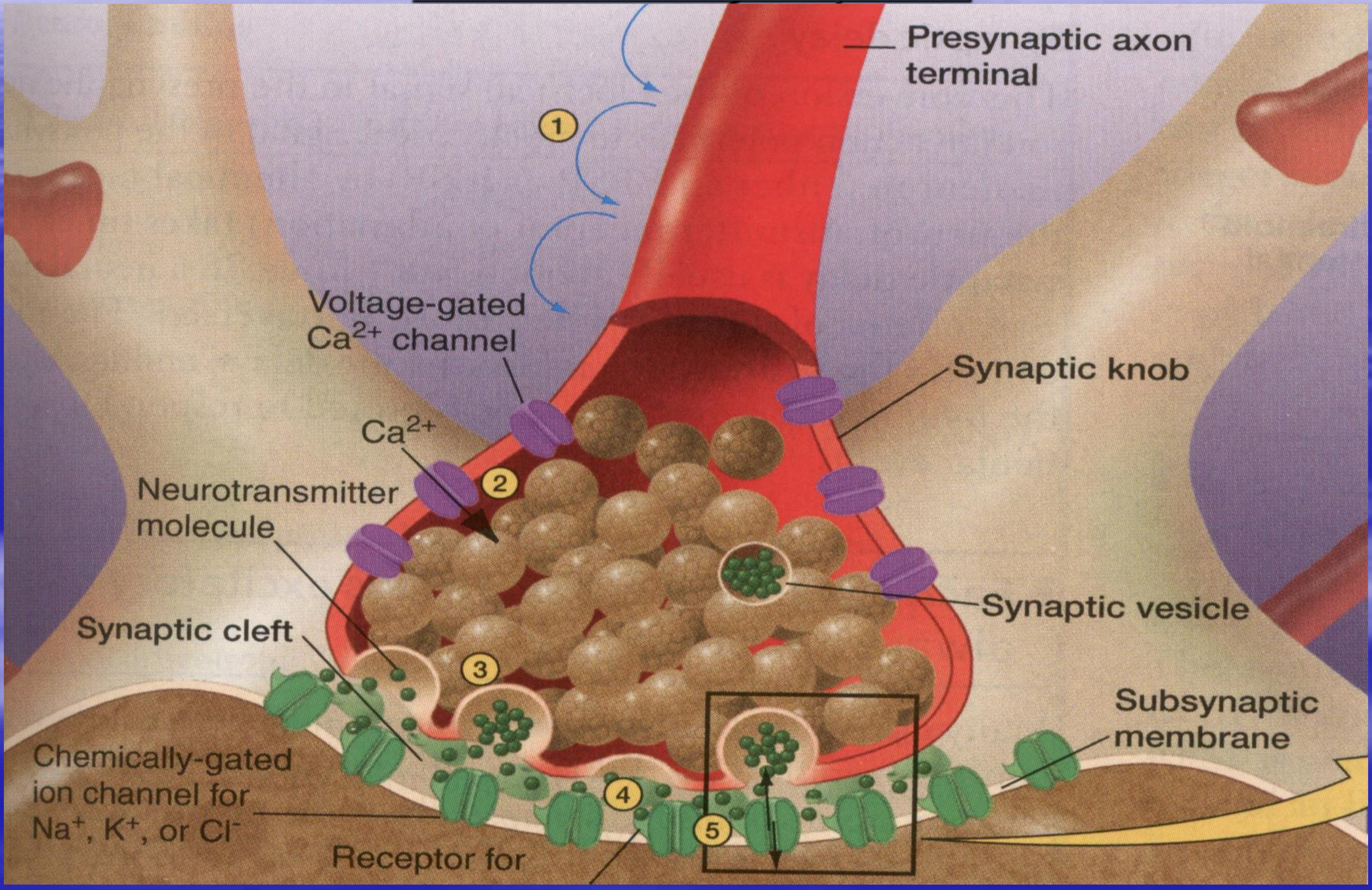
1. Chemical synapse

- Electrical synapse





Mechanism of impulse transmission in chemical synapses



- *It is due to liberation of acetyl choline which is synthesised in the terminal axon.*

Action Potential.ram

- *Nerve impulse causes depolarization of the nerve terminal with opening of the voltage gated Ca^{++} channels.*

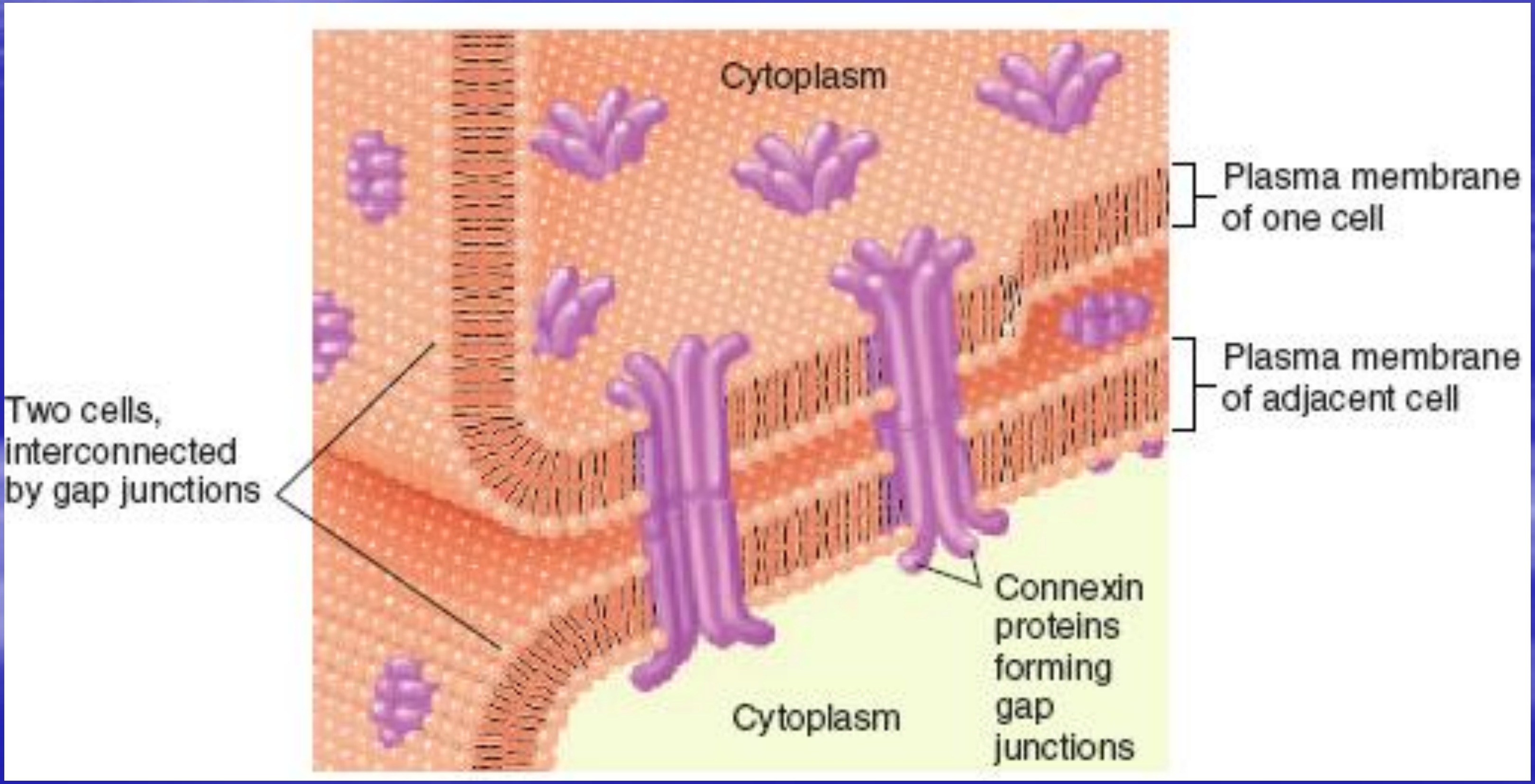
- *This leads to Ca^{++} entry in the nerve terminal & rupture of the vesicles that contain the acetyl choline.*

- *Acetyl choline receptors are present in the post-junctional membrane .*

- *Acetyl choline molecules released in response to nerve impulses bind with the receptors leading to opening of receptor operated ion channels, allowing Na^{+} to flow inside the cell and excite post-synaptic membrane.*

- *Released acetyl choline is rapidly hydrolysed by the enzyme acetyl cholinesterase.*

Electrical: There is low-resistance bridges through which ions pass easily, allowing transmission of the depolarization waves directly from one neuron to another

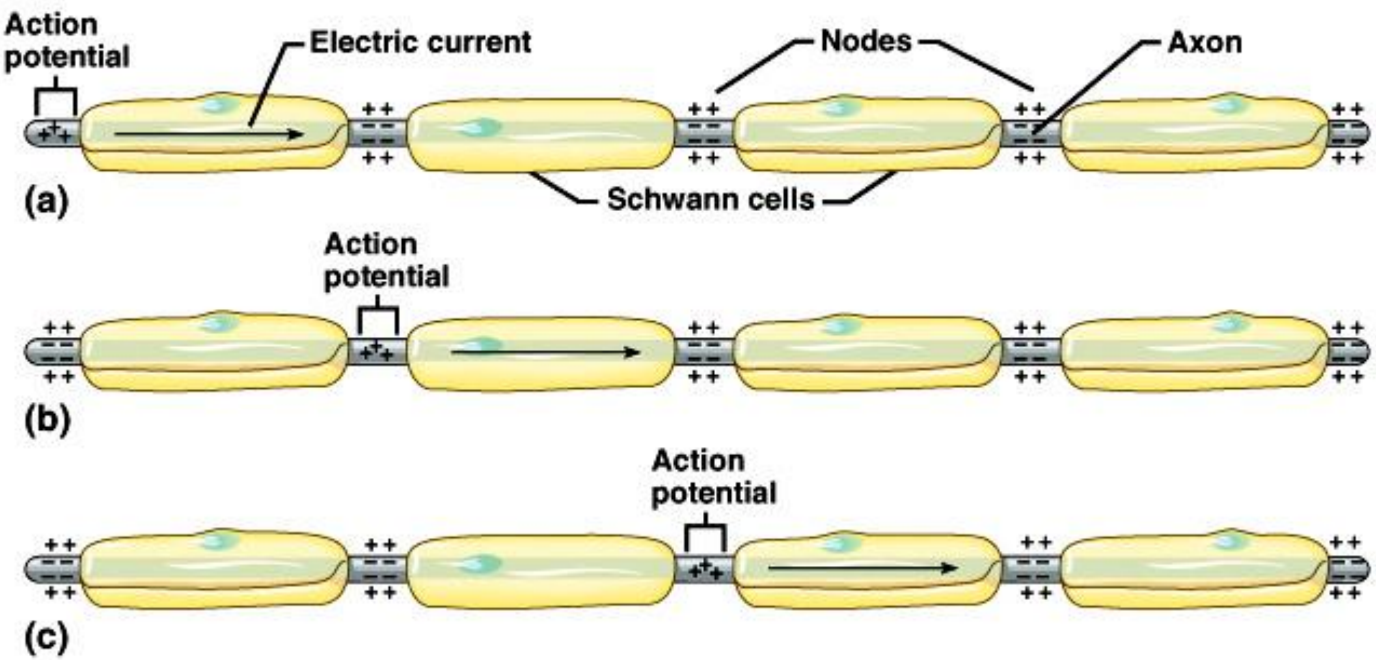


Impulse Conduction

TABLE 10.3 Events Leading to Nerve Impulse Conduction

1. Nerve cell membrane maintains resting potential by diffusion of Na^+ and K^+ down their concentration gradients as the cell pumps them up the gradients.
2. Neurons receive stimulation, causing local potentials, which may sum to reach threshold.
3. Sodium channels in a local region of the membrane open.
4. Sodium ions diffuse inward, depolarizing the membrane.
5. Potassium channels in the membrane open.
6. Potassium ions diffuse outward, repolarizing the membrane.
7. The resulting action potential causes an electric current that stimulates adjacent portions of the membrane.
8. Series of action potentials occurs sequentially along the length of the axon as a nerve impulse.

Saltatory Conduction



Neurotransmitters

TABLE 10.4 Some Neurotransmitters and Representative Actions

Neurotransmitter	Location	Major Actions
Acetylcholine	CNS	Controls skeletal muscle actions
	PNS	Stimulates skeletal muscle contraction at neuromuscular junctions. May excite or inhibit at autonomic nervous system synapses
Biogenic amines		
Norepinephrine	CNS	Creates a sense of well-being; low levels may lead to depression
	PNS	May excite or inhibit autonomic nervous system actions, depending on receptors
Dopamine	CNS	Creates a sense of well-being; deficiency in some brain areas associated with Parkinson disease
	PNS	Limited actions in autonomic nervous system; may excite or inhibit, depending on receptors
Serotonin	CNS	Primarily inhibitory; leads to sleepiness; action is blocked by LSD, enhanced by selective serotonin reuptake inhibitor antidepressant drugs
Histamine	CNS	Release in hypothalamus promotes alertness
Amino acids		
GABA	CNS	Generally inhibitory
Glutamate	CNS	Generally excitatory
Neuropeptides		
Enkephalins, endorphins	CNS	Generally inhibitory; reduce pain by inhibiting substance P release
Substance P	PNS	Excitatory; pain perception
Gases		
Nitric oxide	CNS	May play a role in memory
	PNS	Vasodilation

Neurotransmitters

TABLE 10.7 Events Leading to Neurotransmitter Release

1. Action potential passes along an axon and over the surface of its synaptic knob.
2. Synaptic knob membrane becomes more permeable to calcium ions, and they diffuse inward.
3. In the presence of calcium ions, synaptic vesicles fuse to synaptic knob membrane.
4. Synaptic vesicles release their neurotransmitter by exocytosis into the synaptic cleft.
5. Synaptic vesicles become part of the membrane.
6. The added membrane provides material for endocytotic vesicles.

The Spinal Cord

Functions:

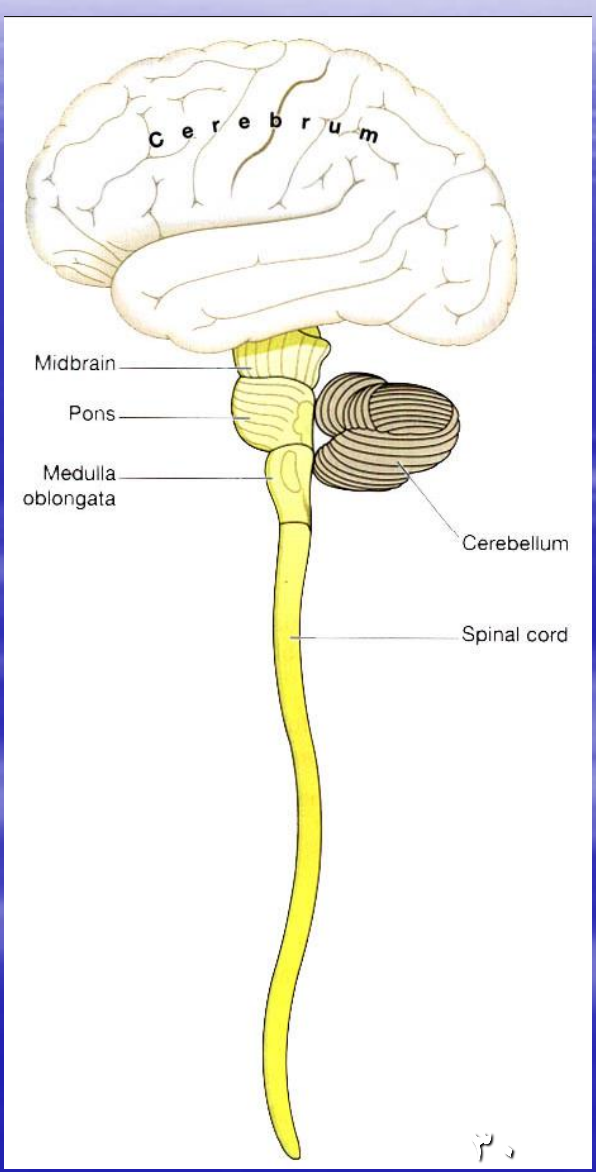
- sensory & motor innervation to entire body below the head

- two-way communication pathway between body and head

- major reflex centre

- **The brain consisted of:**
- cerebrum
- midbrain
- Pons
- medulla oblongata
- cerebellum.

Brain stem



Brain Stem

- **Medulla**

  breathing, blood pressure, swallowing, coughing and vomiting reflexes
- **Pons**

  participates with medulla in regulating breathing

  gets information from cerebral hemispheres to cerebellum
- **Midbrain**

  Controls eye movements

  contains neucli of auditory and visual system

Cerebellum

- Functions:

i. Coordination of movements
ii. Planning and execution of movements
iii. Maintenance of posture
iv. Coordination of head and eye movements

Thalamus and Hypothalamus

- **Thalamus**

  processes all sensory and motor information
- **Hypothalamus**

  contains centers to regulate

1. Body temp
2. Water balance
3. Pituitary gland
4. ADH and oxytocin

The Cerebrum:

- Enables awareness of self, sensation, communication, memory, understanding and movement.

The reflex arc

▪ Is the basic unit of integrated reflex activity

▪consists of:

1- sense organ

2- an afferent neuron

3- one or more synapses that are generally in a central integrating station

4- an efferent neuron

5- an effector.

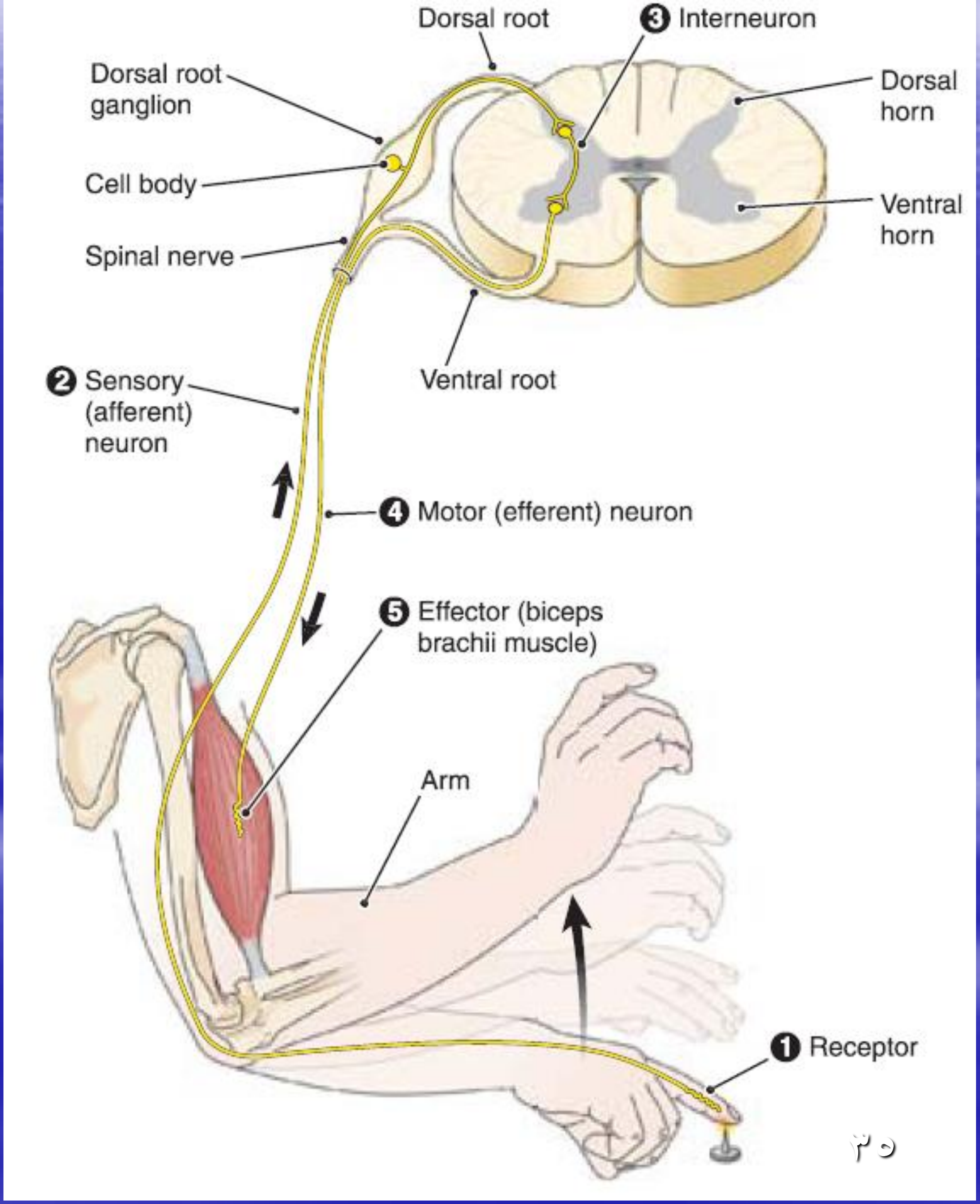


Table 9•2 Components of a Reflex Arc

COMPONENT	FUNCTION
Receptor	End of a dendrite or specialized cell that responds to a stimulus
Sensory neuron	Transmits a nerve impulse toward the CNS
Central nervous system	Coordinates sensory impulses and organizes a response; usually requires interneurons
Motor neuron	Carries impulses away from the CNS toward the effector, a muscle, or a gland
Effector	A muscle or gland outside the CNS that carries out a response

Somatic and Special Senses

The sensory system is divided into 2 sections.

The somatic sensory system is the system responding to information from the skin, muscles and viscera (organs).

The special senses are taste, smell, vision and hearing.

Types of sensory receptors

- Receptor
  - Specialized cell or multicellular structure
  - Collects information about the environment
  - Sends information via afferent pathways to spinal cord and brain.
  - Cerebral cortex then processes information.

Receptors

- Types of receptors
  - Chemoreceptors
    - Sense changes in chemical concentration.
    - Ex: smell and taste
  - Pain receptors (nociceptors)
    - Sense tissue damage from excess stress on tissue.
  - Thermoreceptors
    - Sense temperature change

Receptors

- Types of receptors (cont)
  - Mechanoreceptors
    - Sense changes that deform the receptor
  - Proprioceptors
    - Sense change in tensions of muscles and tendons.
  - Baroreceptors
    - Sense changes in blood pressure in blood vessels.
  - Stretch receptors
    - Sense changes in tissue length (found in lungs).
  - Photoreceptors
    - Sense changes in light intensity.

Sensory Impulses

Stimulation of receptors causes changes in membrane potentials that result in action potentials in sensory fibers.

So all of the receptors essentially do the same thing—they take information about the environment and change it to electrochemical information so it can be processed by the nervous system.

Sensation

When receptors are continually stimulated they become less responsive to the stimulus—**sensory adaptation.**

Example: hot and cold

Receptors that do not adapt: pain, joint, and muscle monitoring receptors

Somatic Senses

The somatic senses consist of sensory receptors associated with skin, muscles, joints and viscera.

Somatic Senses

The somatic senses include:

- Touch and pressure
- Temperature
- Pain
- Stretch

These sensations are processed by various receptors.

the special senses include:

1. Smell
2. Taste
3. Hearing
4. Vision

Special Senses

Smell

- Sensed via olfactory receptors in olfactory organs
  - Chemoreceptors
  - Smell and taste work together
  - 75-80% of flavor comes from sense of smell.

Special Senses

Smell (pathway)

- Olfactory receptors stimulated by substance.
- Nerve impulse travels via fibers running through cribriform plate of ethmoid bone.
- Fibers synapse with neurons located in olfactory bulbs (crista galli of ethmoid bone).
- Impulses are analyzed and travel along olfactory tracts to limbic system (smell may be linked to memory).

Special Senses

Taste

- Taste buds located on surface of tongue (papillae) (also on roof of mouth, linings of cheeks and walls of pharynx.
- Chemoreceptors (taste cells) pick up dissolved substances.

Special Senses

Taste

There are 4 primary taste sensations. So all tastes are combinations of these 4 primary tastes.

1. Sweet (sugars, saccharin, alcohol, and some amino acids)
2. Sour (hydrogen ions)
3. Salty (metal ions)
4. Bitter (alkaloids such as quinine and nicotine)

Special Senses

- Taste is carried by cranial nerves and is processed in the parietal lobe.
  - Impulses travel to medulla oblongata—thalamus—parietal lobe.